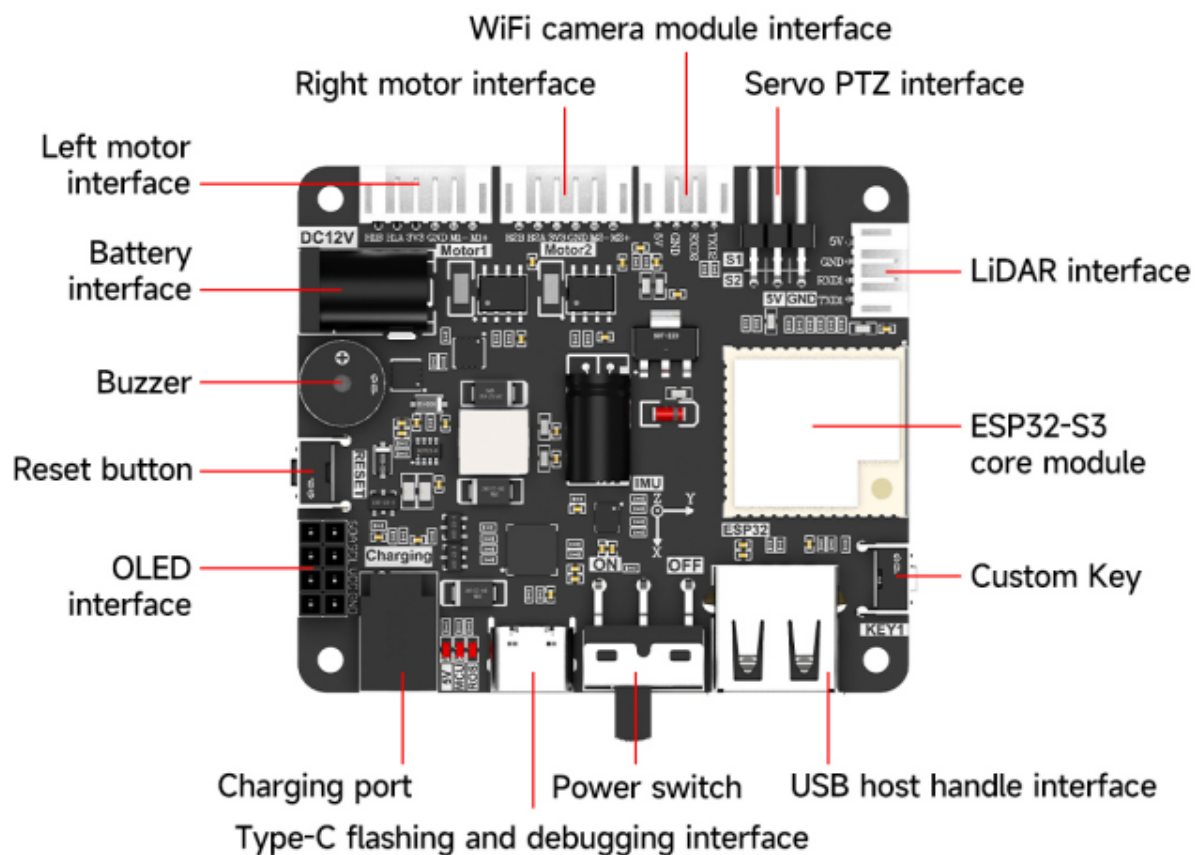


MicroROS-V2 Controller Board

Introduction

The MicroROS-V2 controller board is a ROS 2 robot controller based on the ESP32-S3-WROOM-1U-N4R2 module. The current version features onboard dual-channel encoder motor drivers, dual-channel PWM servo interfaces, a QMI8658B six-axis IMU, a buzzer, and status LEDs, and provides interfaces for the lidar, WiFi image transmission module, OLED, USB Host gamepad, and serial flashing/debugging.

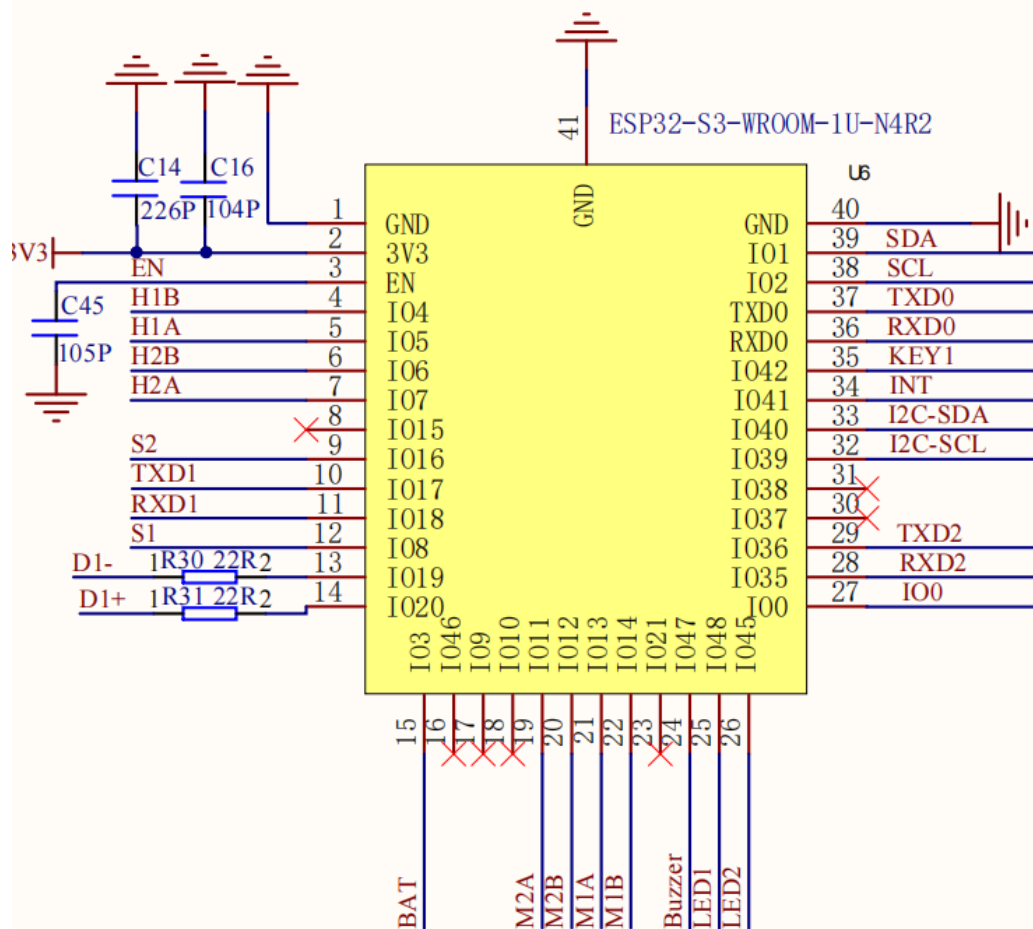
Onboard Resources



Onboard resource	Description
Battery interface	Connects a 12V battery to power the controller board and the robot.
Charging interface	Connects the charging power supply to charge the battery.
Power switch	Controls the power on/off of the entire board.
Reset button	Resets the ESP32-S3 main controller.
Left/right motor interfaces	Connect two DC motors with Hall encoders; the interfaces carry both motor drive and encoder signals.
Servo pan-tilt interface	Provides S1 and S2, two channels of PWM servo control signals, as well as power.
Lidar interface	Connects the T-mini Plus lidar, communicating via UART1.
WiFi image transmission module interface	Connects the companion image transmission/camera module, communicating via UART2.
OLED interface	Connects the I2C OLED display, using an independent I2C1 bus.
USB Host gamepad interface	Connects the USB gamepad receiver; the data lines use the ESP32-S3 native USB D-/D+.
Type-C flashing/debugging interface	Used for firmware flashing, parameter configuration, and UART0 serial debugging.
Custom button	KEY1 user button, active low, whose function can be defined by the firmware.
Buzzer	Onboard passive buzzer for status and low-voltage alarms.
Status LEDs	Onboard blue and green MCU status LEDs.
Six-axis IMU	Onboard QMI8658B accelerometer and gyroscope.
ESP32-S3 module	The main controller of the board, responsible for motors, sensors, communication, and micro-ROS functionality.

GPIO Pin Assignment

ESP32-MCU



In the following, the directions of **TX** and **RX** are defined relative to the ESP32-S3 main controller. For example, **UART1 TX** means data is sent from the ESP32-S3 to the lidar; when wiring, it should be connected to the lidar's **RX**.

Motors and Encoders

Interface/Function	Schematic net name	ESP32-S3 GPIO	MCU direction
Left motor M1 drive A	M1A	GPIO13	Output
Left motor M1 drive B	M1B	GPIO14	Output
Left motor M1 encoder A	H1A	GPIO5	Input
Left motor M1 encoder B	H1B	GPIO4	Input
Right motor M2 drive A	M2A	GPIO11	Output
Right motor M2 drive B	M2B	GPIO12	Output
Right motor M2 encoder A	H2A	GPIO7	Input
Right motor M2 encoder B	H2B	GPIO6	Input

Servos, Display, and IMU

Interface/Function	Schematic net name	ESP32-S3 GPIO	Description
Servo S1	S1	GPIO8	PWM output; used by the firmware for the horizontal (pan) direction
Servo S2	S2	GPIO16	PWM output; used by the firmware for the pitch direction
OLED data	SDA	GPIO1	I2C1 SDA
OLED clock	SCL	GPIO2	I2C1 SCL
IMU data	I2C-SDA	GPIO40	I2C0 SDA
IMU clock	I2C-SCL	GPIO39	I2C0 SCL
IMU interrupt	INT	GPIO41	IMU interrupt input

OLED and IMU use two independent I2C buses and must not be wired to the same set of SDA/SCL pins.

Serial Port and USB

Interface/Function	Schematic net name	ESP32-S3 GPIO	Connection
Lidar UART1 TX	TXD1	GPIO17	ESP32-S3 TX -> lidar RX
Lidar UART1 RX	RXD1	GPIO18	ESP32-S3 RX <- lidar TX
WiFi image transmission UART2 TX	TXD2	GPIO36	ESP32-S3 TX -> image transmission module RX
WiFi image transmission UART2 RX	RXD2	GPIO35	ESP32-S3 RX <- image transmission module TX
Flashing/debugging UART0 TX	TXD0	GPIO43	ESP32-S3 TX -> USB-to-serial RX
Flashing/debugging UART0 RX	RXD0	GPIO44	ESP32-S3 RX <- USB-to-serial TX
USB Host D-	D1-	GPIO19	Native USB data line D-
USB Host D+	D1+	GPIO20	Native USB data line D+

Buttons, Detection, and Status Output

Interface/Function	Schematic net name	ESP32-S3 GPIO	Description
Battery voltage detection	BAT	GPIO3	ADC input
Custom button	KEY1	GPIO42	Digital input, active low
Download/BOOT control	IO0	GPIO0	Boot configuration pin
Buzzer	Buzzer	GPIO47	PWM output, drives the passive buzzer
Blue status LED	LED1	GPIO48	Digital output, lights up on high level
Green status LED	LED2	GPIO45	Digital output, lights up on high level

Wiring Notes

- The logic level of the ESP32-S3 GPIOs is 3.3V, and the GPIOs are not 5V tolerant; do not connect the 5V power pin of an interface directly to a GPIO signal pin.
- UART must be cross-connected: connect this board's **TX** to the peripheral's **RX**, and this board's **RX** to the peripheral's **TX**, and make sure the two ends share a common ground.
- The numbering of motors M1/M2 follows the schematic net names; the left/right orientation depends on the current firmware configuration. When adjusting the mechanical mounting orientation, do not swap the encoder A/B definitions.
- GPIO0 is the boot configuration pin; during startup or reset, do not hold it externally low for a long time, otherwise the board may enter download mode.
- Before connecting or disconnecting motors, servos, the lidar, and image transmission modules, it is recommended to turn off the controller board power first.